

Historical-Simulation VaR Stress Engine — Cheatsheet

For the Just role: extend a Python stress engine to bonds, credit spreads, etc. · revise out loud

★ MUST-KNOW COLD

1. **Historical-simulation VaR** = re-apply each past day's risk-factor move to *today's* portfolio, **revalue**, collect the **P&L distribution**, take the tail loss. No distributional assumption — the history is the distribution.
2. **99.5% / 1-year is the Solvency II SCR** market-risk calibration → this engine likely feeds **capital**.
3. **VaR(99.5%)** = $-\text{sorted_pnl}[\text{floor}((1-0.995) \cdot N)]$ — the loss at the 0.5% -worst scenario.
4. **Extending to a new instrument** = add (a) a pricing function and (b) its risk factors to the scenario set — *not* an engine rewrite.
5. **Shock conventions differ by factor**: rates & credit spreads → **absolute (bp)**; equities & FX → **relative (%/log)**. Getting this right is the classic error.
6. **Full revaluation > sensitivity** for a 99.5% tail because of **convexity** (big moves); sensitivity (DV01/CS01) is the fast approximation.
7. **Bond risk factors** = the rate curve + the credit spread. **CS01** = P&L per 1bp spread; **DV01** = per 1bp rate.
8. **ES (Expected Shortfall)** = average loss *beyond* VaR — tail-aware; FRTB & many internal models prefer it.

1. The algorithm (own these steps)

1. **Lookback window**: N historical days ($\approx 1-3$ yrs; 250–750).
2. **Scenarios**: for each day, the **change in every risk factor** (Δrate , Δspread , Δequity , ΔFX).
3. **Apply to today**: shock *today's* market inputs by that day's move.
4. **Revalue**: $\text{P\&L}_i = V(\text{shocked}_i) - V(\text{base})$.
5. **Distribution**: the N P&Ls. **Sort ascending**.
6. **Tail**: $\text{VaR}(\text{conf}) = -\text{sorted_pnl}[\text{floor}((1-\text{conf}) \cdot N)]$ (loss as a positive number).

$\text{VaR}(99.5\%) = -\text{sorted_pnl}[\text{floor}(0.005 \cdot N)]$ $\text{ES} = -\text{mean}(\text{losses beyond VaR})$

Example: $N=250 \rightarrow \text{floor}(0.005 \cdot 250)=1 \rightarrow$ VaR is the **2nd-worst** P&L. (250 scenarios is thin for 99.5% — real engines use longer windows or scale.)

2. Risk factors & shock conventions

- **Rates / credit spreads** → **ABSOLUTE bp moves**: $s_{\text{new}} = s_{\text{today}} + \Delta s_{\text{hist}}$.
- **Equities / FX** → **RELATIVE returns**: $S_{\text{new}} = S_{\text{today}} \cdot (1 + r_{\text{hist}})$ (or log).
- **Why it matters**: applying a 2008 *relative* rate move to today's low rates (or vice-versa) gives nonsense. Match the convention to the factor.

3. Revaluation: full reval vs sensitivity

- **Full revaluation**: reprice the instrument under each shocked market. **Accurate, captures convexity & cross-effects**. Cost: pricing $\times N$ scenarios.
- **Sensitivity (Taylor)**: $\Delta P \approx -\text{DV01} \cdot \Delta y - \text{CS01} \cdot \Delta s + \frac{1}{2} \cdot \text{Conv} \cdot \Delta y^2$. **Fast**, but misses higher-order terms on big shocks.
- **For 99.5% tail / bonds** → **prefer full reval** (convexity bites). Use sensitivity for speed or sanity checks.

4. Extending to bonds & credit (the actual task)

- **Bond price (full reval)**: $P = \sum CF_t \cdot \exp(-(z_t + s) \cdot t)$ — discount on **rate curve z + credit spread s** .
- **Risk factors to add**: the **rate curve points** and the **credit-spread (by rating/sector/issuer)** — both must enter the historical scenario set.
- **CS01** = $P(s) - P(s+1\text{bp}) \approx \text{spread-duration} \times \text{price} \times 1\text{bp}$. **DV01** = $P(y) - P(y+1\text{bp})$.
- **CDS / hazard-rate instruments**: value via **survival probabilities** $Q(t) = \exp(-\int \lambda)$; spread shocks move the hazard/credit curve.
- **Mapping/proxy**: a bond with no own spread history → **proxy** to a rating/sector spread series.
- **Watch**: FX for non-base-currency bonds; pull-to-par; coupon/accrual; callability (optionality → needs a model, not just discounting).

5. Engine architecture / extensibility (they'll probe your design)

- **Instrument interface** with `revalue(market)`; each instrument **declares its risk-factor dependencies**.
- **Scenario engine**: historical Δ risk-factors → build `shocked_market` → revalue all instruments → **portfolio P&L vector** → percentile.
- **Add an instrument = new pricer class + its risk-factor series**; the engine stays generic (it never knows instrument internals).
- **Decisions to raise**: full-reval vs sensitivity, absolute-vs-relative per factor, risk-factor **proxying**, curve interpolation, **vectorisation** (numpy: scenarios $\times$ factors matrix), caching the base valuation.

6. Horizon, scaling, and cousins

- **Holding period**: 1-day VaR → scale to h-day by $\sqrt{h}$ (i.i.d. assumption), or use **overlapping h-day returns**. **SCR = 1-year, 99.5%**.
- **Filtered Historical Simulation (FHS)**: scale historical moves by **current vs historical volatility** (e.g. EWMA) so the engine is responsive — good depth to mention.
- **Anti-procyclicality (APC)**: floors/buffers so capital doesn't collapse in calm and spike in stress.
- **Backtesting**: count exceptions vs expected (e.g. ~ 2.5 breaches/yr at 99% daily) — regulatory traffic-light.

7. Likely questions → answers

- **"Add bonds to the engine?"** → a bond pricer consuming **rate curve + credit spread**; add the spread series to scenarios; **full-reval** each scenario.
- **"How handle credit spreads?"** → CS01 / spread curve, **absolute bp** shocks, **proxy** missing names; CDS via hazard rates.
- **"Write 99.5% historical VaR from a P&L vector."** → `sort, idx=floor(0.005 * N), -sorted[idx]` (mind the off-by-one & sign).
- **"Full reval vs sensitivity trade-off?"** → accuracy/convexity vs speed; tail favours full reval.
- **"Design so a new instrument is easy?"** → instrument/pricer abstraction + risk-factor registry; engine generic.
- **"VaR vs ES?"** → ES averages the tail beyond VaR; coherent; FRTB uses ES.

8. Gotchas (say these unprompted — signals seniority)

- **Percentile indexing / off-by-one** and **P&L sign** (loss = negative P&L).
- **Absolute vs relative** shocks per factor.
- **Window length vs confidence**: 99.5% needs a long history (or scaling) — 250 days is thin.
- **Missing risk-factor history** → **proxy**, and **stale/holiday data alignment** across factors.
- **Ghost effect**: a big move drops out of the window after N days → VaR jumps; FHS mitigates.

Formula quick-ref

Bond (full reval)	$P = \sum CF_t \cdot \exp(-(z_t + s) \cdot t)$	$s = \text{spread_bp} / 1e4$
DV01	$P(y) - P(y+1bp)$	$CS01 = P(s) - P(s+1bp)$
Scenario P&L_i	$V(\text{shock today by day-i moves}) - V(\text{base})$	
Hist VaR(conf)	$-\text{sorted_pnl}[\text{floor}((1-\text{conf}) \cdot N)]$	
ES(conf)	$-\text{mean}(\text{pnls below the VaR threshold})$	
h-day scaling	$\text{VaR}_h = \text{VaR}_{1day} \cdot \sqrt{h}$	SCR = 99.5%, 1-year
Rates/spreads	absolute bp shock	Equity/FX relative % shock

Practise the engine end-to-end in `hist_var_drill.py` (fill-in, graded): bond pricer → DV01/CS01 → P&L vector by full reval → 99.5% VaR. Key `hist_var_drill_SOLUTIONS.py`.